

Introduction to Computer Simulation II

Homework 7

Due: Tuesday, 02 June 2026

13. Cluster estimator for the correlation function of the 2D Ising model

In the high-temperature phase of the Ising model the correlation function $G(\vec{x}_i, \vec{x}_j) = \langle s_{\vec{x}_i} s_{\vec{x}_j} \rangle$ can be estimated by the mean value of $s_{\vec{x}_i} s_{\vec{x}_j}$, where for periodic boundary conditions (translational invariance) also all $\vec{x}_i, \vec{x}_j$ with the same (vectorial) distance $\vec{R}_{ij} \equiv \vec{x}_i - \vec{x}_j$ can be included. Alternatively, $G(\vec{x}_i, \vec{x}_j)$ can be determined by the cluster estimator $\hat{G}(\vec{x}_i, \vec{x}_j) = \frac{V}{|C|} \Theta_C(\vec{x}_i) \Theta_C(\vec{x}_j)$, where $|C|$ is the number of spins in the cluster and $\Theta_C(\vec{x}_i)$ the characteristic function ($= 1$ if $\vec{x}_i$ is in the cluster and 0 otherwise).

Implement both measurement prescriptions for the 2D Ising model on a quadratic lattice of linear extent $L = 64$ and 128 with periodic boundary conditions for the special case that $\vec{R}_{ij} = \vec{x}_i - \vec{x}_j$ points along the x -axis. Use for the simulations the single-cluster algorithm of problem 4 and compare the accuracy of the two measurement prescriptions.

For sufficiently large distances $|\vec{R}_{ij}|$ the correlation function decays exponentially, $G \propto \exp(-|\vec{R}_{ij}|/\xi)$, where $\xi = \xi(\beta)$ is the (spatial) correlation length. Estimate ξ for $\beta = 0.42, 0.40, 0.35$, and 0.30 .

14. Statistical error of histograms

Simulate the 2D Ising model for a 16×16 lattice with periodic boundary conditions at $\beta_c = \ln(1 + \sqrt{2})/2 = 0.440686\dots$ with the Metropolis algorithm (5000 sweeps for thermalisation, $2^{16} = 65536$ sweeps for measurements) and determine the energy and the energy histogram as unnormalized number histogram $H(E)$. For both quantities record 16 blocks, from which the statistical errors can be obtained. Estimate for the energy the integrated autocorrelations time $\tau_{\text{int},E}$ and compare for the number histogram the statistical error with the asymptotic result $\sqrt{H(E)}$ for uncorrelated data. Is the measured statistical error larger by a factor of $\sqrt{2\tau_{\text{int},E}}$?